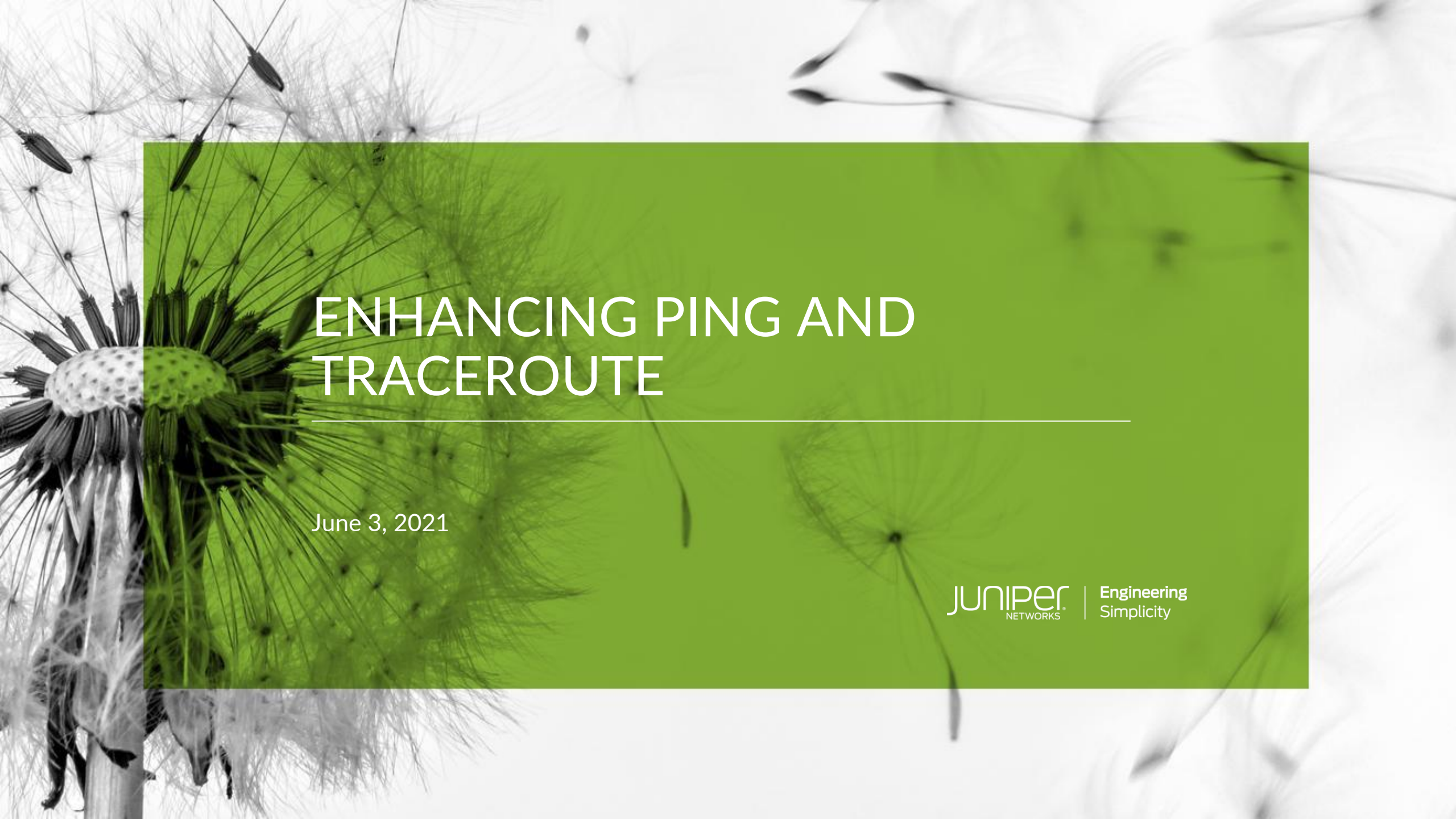


ENHANCING PING AND TRACEROUTE

June 3, 2021

JUNIPER NETWORKS | Engineering Simplicity

Credits

- Developers (Harvey Mudd College)
  - Ishaan Gandhi
  - Andreas Roeseler
  - Spencer Lang
  - Nick Ludwig
- Mentors
  - Ron Bonica (Juniper Network)
  - Zach Dodds (Harvey Mudd College)

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THE IP OAM TOOLKIT

- PING and TRACEROUTE are the most commonly use tools in the IP OAM Toolkit
- What do they do?
- How do they work?
- What can't they do?
- ***How can they be enhanced?***

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PING

WHAT DOES IT DO?

- Test the liveliness of a ***reachable*** interface
- Test the liveliness of a ***reachable*** node
  - By testing the liveliness of one of its ***reachable*** interfaces

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HOW DOES IT WORK?

- A probing node sends an ICMP Echo to a probed interface
- The probed interface sends an ICMP Echo Reply to the probing node
- Nerd Notes
  - The ICMP Echo may enter the probed node through any of its interfaces
  - The ICMP Echo reply message may leave the probed node through any of its interfaces
  - There is no guarantee that that either ICMP message traverses the probed interface

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WHAT CAN'T IT DO?

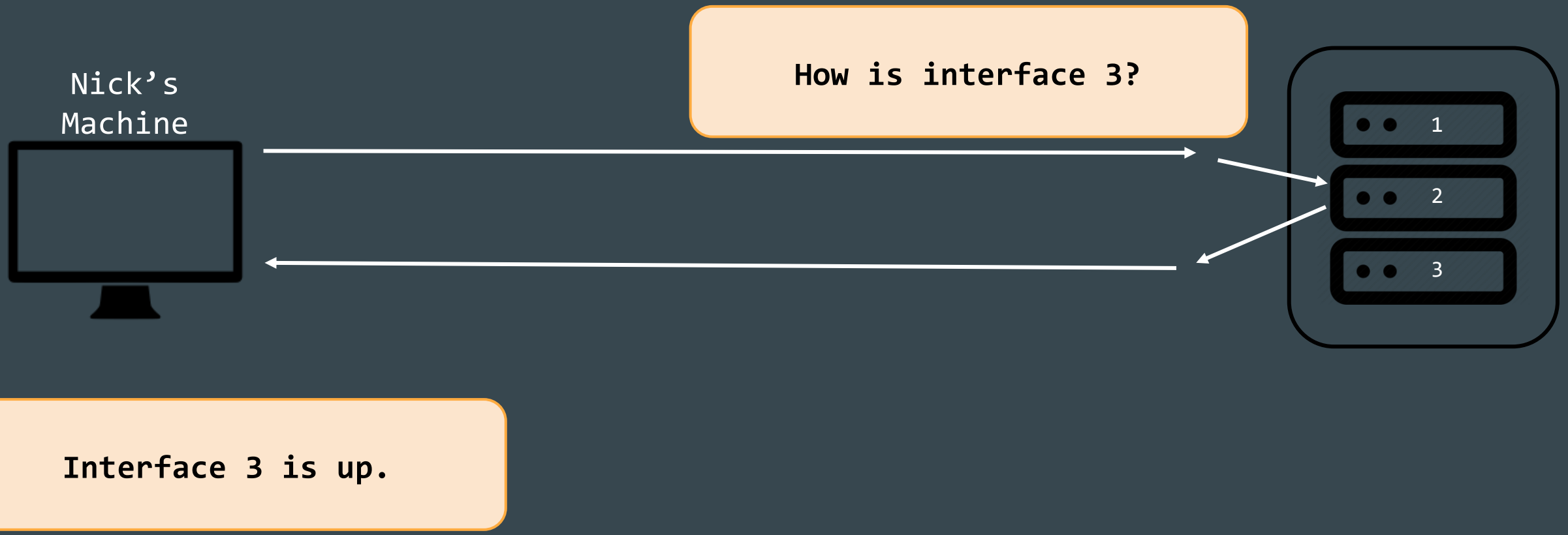
- Test the liveliness of ***less-than-reachable*** interface
- Examples of ***less-than-reachable*** interfaces
  - IPv4 unnumbered
  - IPv6 unicast with narrowly scoped address (link-local, ULA)
  - Any interface to which the probing node lacks a route

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HOW CAN IT BE ENHANCED? [RFC 8335]

- A **probing** node sends an ICMP Extended Echo Request to a **proxy** interface
  - ICMP Extended Echo Request identifies the **probed** interface
- The proxy interface sends an Extended ICMP Echo Reply to the probing node
  - The extended ICMP Echo Reply reports the liveliness of the probed interface

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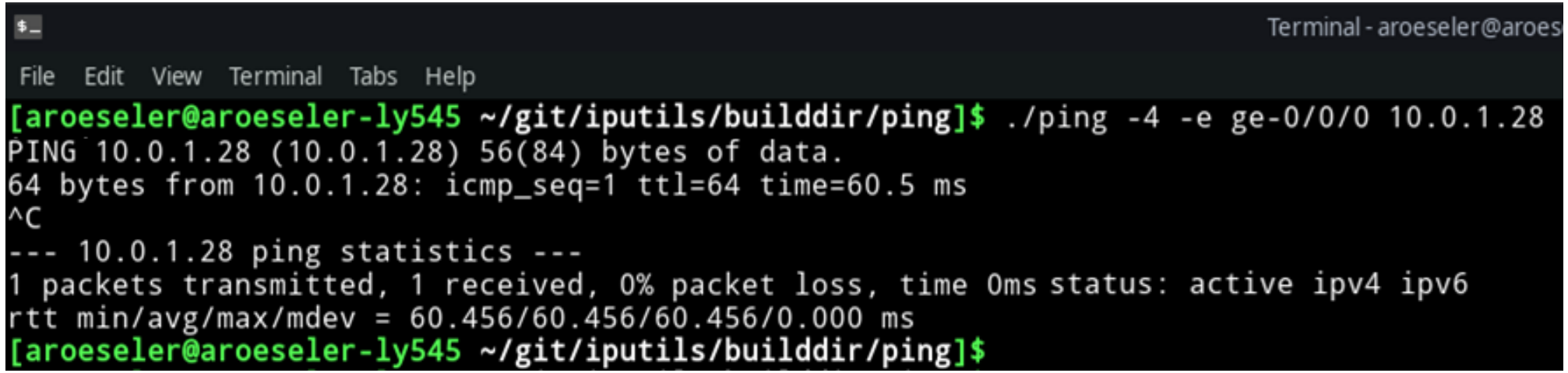
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RFC 8335 NERD NOTES

- The proxy interface can be different from the probed interface
  - And is different in most cases
- The proxy interface must be ***reachable*** from the probing node
- The proxy interface must reside on one of the following
  - The same node as the probed interface
  - A node that is directly connected to the probed interface

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Example Usage



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SECURITY CONSIDERATIONS

- Not enabled by default
- Accessible from specified source addresses only

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IMPLEMENTATIONS

- JUNOS 20.3R1
- LINUX Kernel(5.13)
- LINUX IP Utils Ping (in progress)
- Wireshark (3.5)
- TCPDUMP (in progress)

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TRACEROUTE

WHAT DOES IT DO?

- Elicit feedback from each ***node*** on the delivery path between a probing interface and a destination interface
- Identify ***nodes*** along the delivery path

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HOW DOES IT WORK?

- A probing node sends a series of UDP packets to a destination interface
  - Sets the TTL to 1 on the first packet, so that it expires on the first node along the delivery path
  - Increments the TTL on each subsequent, so that it expires on the next node along the delivery path
- When a packet expires on a node along that delivery path, that node sends a ICMP Time Expired message to the probing node

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NERD NOTES

- Regarding the UDP probe messages
  - By default, on the first packet, the probing node sets the UDP destination port to 33434
  - Increments UDP destination port on each subsequent packet
- Regarding the ICMP Time Expired message
  - The source address may not identify the interface upon which the UDP probe message arrived
    - IPv4: Identifies the interface through which the ICMP message left the reporting node
    - IPv6: It's complicated. See RFC 6724.

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WHAT CAN'T IT DO?

- Identify ***interfaces*** along the delivery path

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HOW CAN IT BE ENHANCED? [RFC 5837]

- UDP Probe message is unchanged
- ICMP Time Expired message can contain extensions that identify
  - Interface upon which the UDP probe message arrived
  - Interface through which the message would have been routed had the TTL not expired
  - Attributes of those interfaces (name, IP address, MTU)

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Example Usage

```
[ishaangandhi@Ishaans-MacBook-Pro-5.local ~/Desktop]$ traceroute -v cs.hmc.edu
traceroute to cs.hmc.edu (134.173.42.100), 64 hops max, 52 byte packets
 1  192.168.0.1 (192.168.0.1)  4.885 ms  3.098 ms  3.378 ms
 2  10.79.240.1 (10.79.240.1)  16.694 ms  9.498 ms  15.203 ms
 3  100.127.5.90 (100.127.5.90)  17.656 ms  12.379 ms  15.827 ms
 4  100.120.102.34 (100.120.102.34)  13.698 ms  15.453 ms  10.932 ms
 5  68.1.4.252 (68.1.4.252)  15.853 ms  19.949 ms  28.248 ms
    Arrival interface:
        name: en2
        address: 38.1.5.49
        MTU: 1500
 6  100ge16-2.core1.lax1.he.net (216.218.224.117)  17.620 ms  18.293 ms  17.679 ms
 7  100ge14-1.core1.lax2.he.net (72.52.92.122)  17.727 ms  18.212 ms  22.573 ms
 8  65.19.156.114 (65.19.156.114)  19.951 ms
    216.218.223.26 (216.218.223.26)  21.678 ms  20.780 ms
 9  130.152.184.99 (130.152.184.99)  23.593 ms  17.517 ms
    130.152.184.162 (130.152.184.162)  19.027 ms
10  hmc-a.router.claremont.edu (134.173.253.23)  24.651 ms
    130.152.184.162 (130.152.184.162)  22.136 ms  19.224 ms
11  * * hmc-a.router.claremont.edu (134.173.253.23)  23.770 ms
12  * knuth.cs.hmc.edu (134.173.42.100)  18.032 ms  19.201 ms
    Arrival interface:
        if index: 2
        address: 130.122.104.62
        MTU: 1500
```

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SECURITY CONSIDERATIONS

- Not enabled by default
- Accessible from specified source addresses only

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IMPLEMENTATIONS

- JUNOS (in progress, EX first)
- LINUX Kernel (in progress)
- LINUX IP Utils Traceroute (in progress)
- Wireshark (3.5)

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THANK YOU

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Ron Bonica, Ishaan Gandhi, Spencer Lang, Nick Ludwig, Andreas Roeseler